Start2impact ·

BioMarket

Advanced Analytics

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Final Project · 2024

Context & Sustainability

Eco-friendly brand positioning and dataset overview

Organic Food

Certified bio products, short supply chain, low-emission logistics

Clean Beauty

Cruelty-free cosmetics, compostable packaging, microplastic-free

Eco-Electronics

Energy-efficient electronics, refurbished & circular economy

1000
Sales Transactions

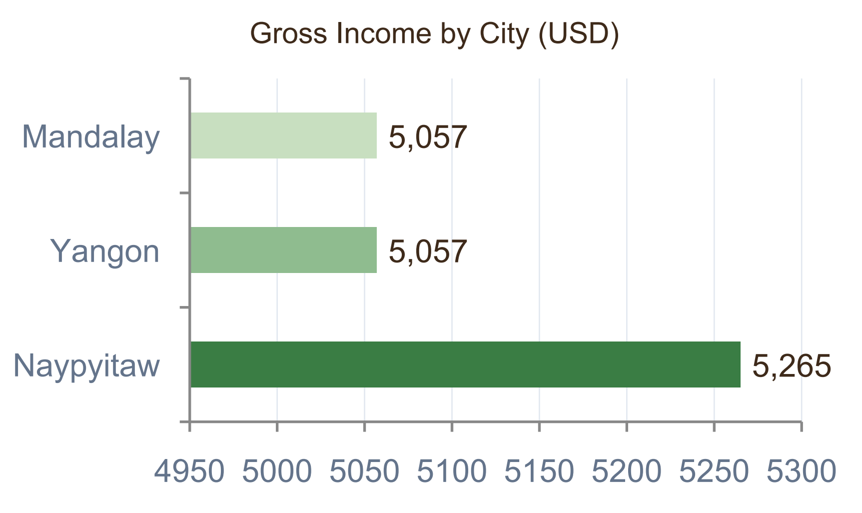
4000
Apples Classified

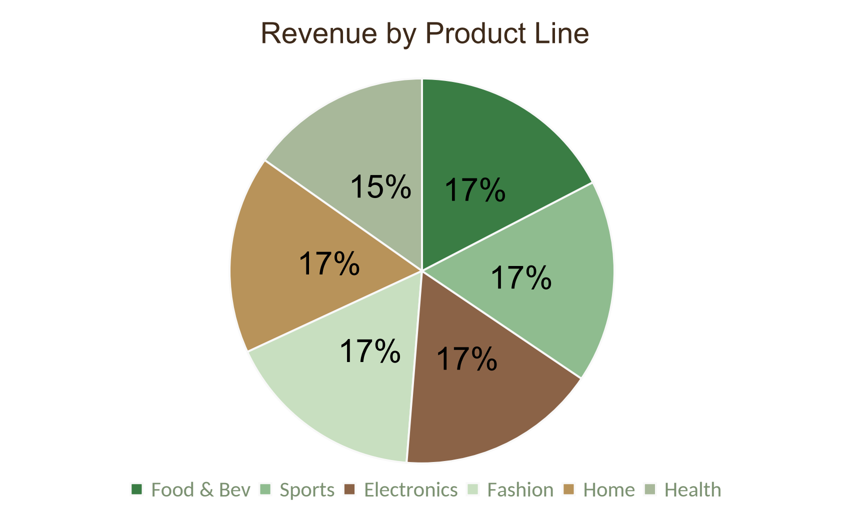
6
Product Lines

3
Cities Analysed

Sales Analysis

Most profitable cities · Top categories · Gender impact on purchases





+7.3%

Naypyitaw vs other cities

Female +8%

Female vs male purchases

Food & Bev

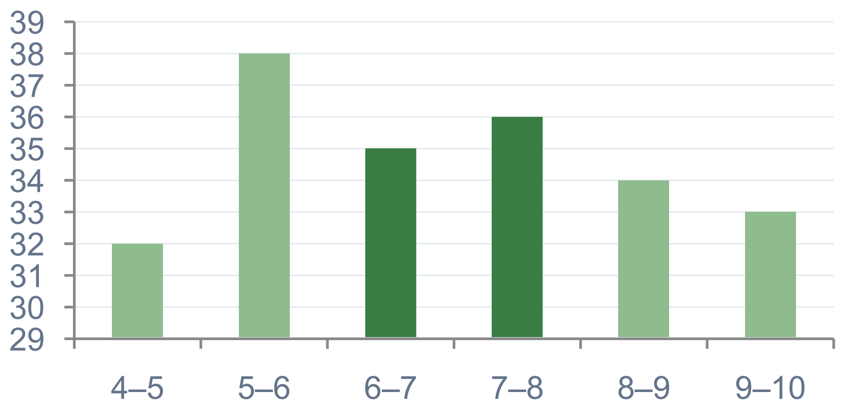
Top revenue category

Gross Income & Rating Distribution

Mean · Median · Mode · Std Dev — and why they tell different stories

6.97	7.00	6.00	1.72	+0.89
Mean	**Median**	**Mode**	**Std Deviation**	**Income Skewness**
avg rating across 1,000 tx	*50th percentile*	*most frequent rating*	*spread: 5.25 → 8.69*	*right tail — few large sales*

Rating distribution — approx. uniform (no skewness)



Gross Income — a different story

Skewness = +0.89 (right tail)

Most transactions have low gross income

Few high-value purchases pull the tail right

Many small receipts, rare large ones

Rating ≈ mean (6.97) ≈ median (7.00) ≈ mode (6.00)

ML: Profitability Prediction

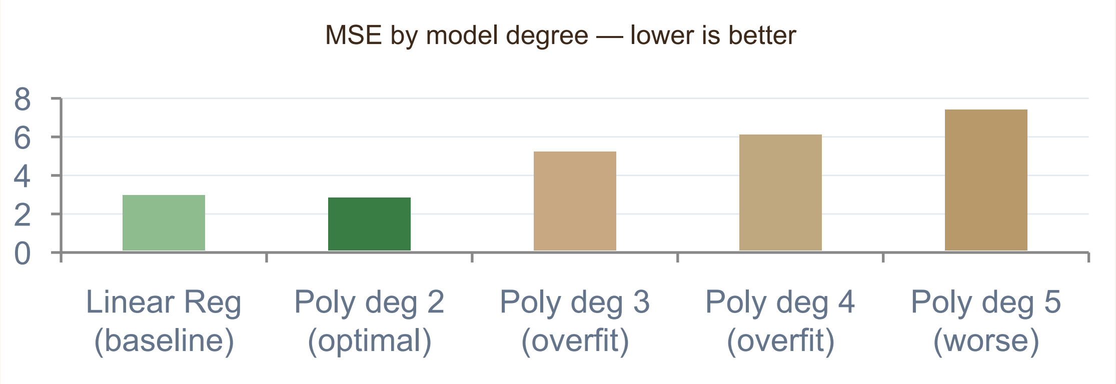
Linear Regression · Polynomial Regression · Rating as profitability proxy

1 **Feature Engineering**
Dropped: Invoice ID, Tax 5%, Total, Date, Time, cogs, margin

2 **Encoding & Scaling**
OneHotEncoder (6 cat. cols) + StandardScaler on Unit price & gross income

3 **Train / Test Split**
80% training · 20% test random_state=42

4 **Linear Regression**
Baseline model — simplest choice, most stable



Key Insight

- **Poly deg 2 = optimal**
- MSE: 2.98 → 2.85 (−0.13)
  Deg 3+ overfits: categorical features don't benefit from poly expansion

Apple Quality: Vertical Analysis

Logistic Regression · Decision Tree · K-Means Clustering

Logistic Regression

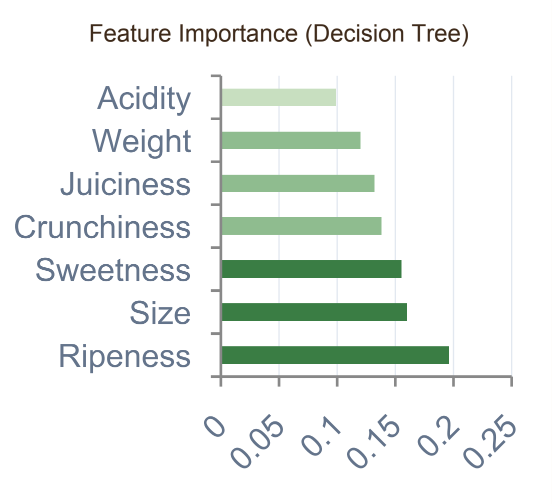
	P:Bad	P:Good
R:Bad	297 ✓	104 ✗
R:Good	87 ✗	312 ✓

LR: 75% accuracy | AUC = 0.82

Decision Tree (optimised)

	P:Bad	P:Good
R:Bad	330 ✓	71 ✗
R:Good	81 ✗	318 ✓

DT: 82% accuracy | F1: 0.82/0.81

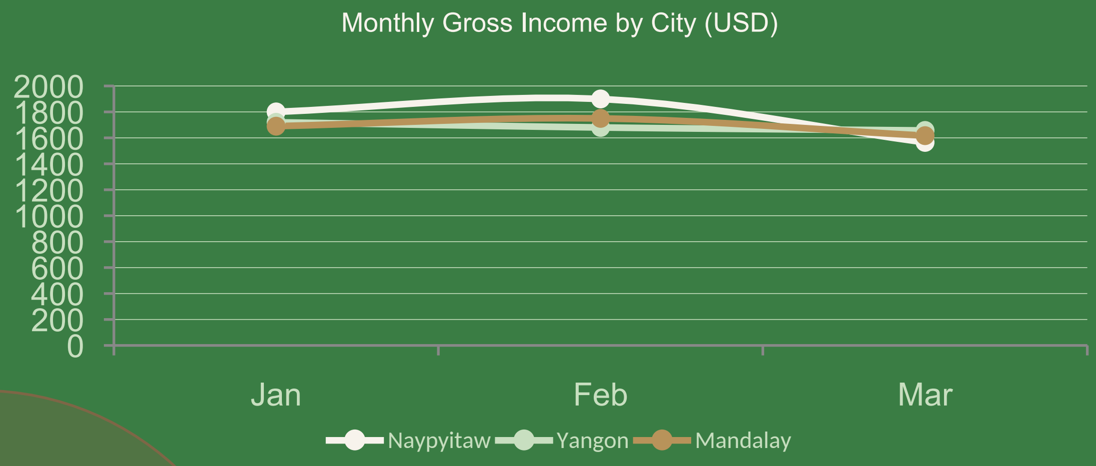


K-Means Clustering

2 clusters: bad vs good — test apple → Cluster 1 (bad)
3 clusters: adds intermediate group (Cluster 1 ≈ 50/50)
PCA visualisation: heavy overlap between clusters

Time Series & Conclusions

Gross income over time · Stationarity vs growth · Strategic recommendations



Linear Regression limitation: gross income oscillates with no clear growth trend (stationary behaviour). A straight line only predicts around the mean — MSE/MAE appear acceptable but the model learns nothing useful about the temporal pattern.

Naypyitaw #1

Most profitable branch
→ prioritise investments

Decision Tree 82%

Best apple classifier
(Ripeness = top feature)

Food & Bev Top

Priority bio category
→ expand assortment

Sources & References

Datasets, tools and project resources

supermarket_sales.csv

Primary Dataset

Sales transactions dataset — 1,000 rows across 3 branches (Yangon, Mandalay, Naypyitaw). Contains product lines, payment methods, gross income and customer ratings. Encoding: windows-1254.

apple_quality.csv

Primary Dataset

Apple quality dataset — 4,000 records. Features: Size, Weight, Sweetness, Crunchiness, Juiciness, Ripeness, Acidity. Binary label: good/bad. One anomalous Acidity row removed during cleaning.

Google Colab — Python Analysis

(click to open → https://colab.research.google.com/drive/1nJ9OjQoJZ3CKJlw-q_rg_xizSV-EuQX7

Notebook

Full analysis notebook: EDA, feature engineering, linear & polynomial regression, logistic regression (GridSearchCV, ROC/AUC), decision tree, K-Means clustering (PCA viz) and time series modelling.